

Acoustic Diode with Nonlinear Lattices

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Abstract

This project investigates non-reciprocal wave propagation in a nonlinear spring-mass lattice, functioning as an acoustic diode. A localized defect mode is introduced by embedding a light mass in an otherwise uniform monoatomic chain, excited with harmonic forcing. Light mass defect induces an exponentially localised mode. Non-linearity from springs leads to amplitude-dependent threshold behavior in wave propagation. Wave transmission occurs only beyond a threshold amplitude, and strong asymmetry is observed between forward and reverse configurations. The resulting transmission ratio is of the order 10^5 , demonstrating diode-like behavior. Such systems have potential applications in wave-based logic, signal filtering, vibration isolation, and acoustic sensing.

Contents

1	Introduction	2
2	System description	2
3	Methodology	3
4	Results and Discussion	4
5	Conclusion and Future Work	7
	Appendix: Matlab Code	8

1 Introduction

Wave propagation control in mechanical systems is a growing area of interest, especially in the context of engineered materials. In traditional linear systems, such as a monoatomic lattice, wave propagation is symmetric and bidirectional, i.e., energy can travel equally in both directions. While this is acceptable in many applications, it poses limitations when unidirectional or non-reciprocal behavior is desired, for example, in vibration isolation, noise control, or wave-based logic circuits.

To address this, researchers¹ have explored how the interplay of nonlinearity, periodicity, and structural asymmetry in granular crystals results in a novel type of switching and rectification devices. One such concept is the acoustic diode, a mechanical analog of the electrical diode, which permits wave transmission in one direction while suppressing it in the opposite direction. Acoustic diodes achieve non-reciprocal behavior through physical configuration and nonlinear dynamic response.

Here, we investigate the acoustic diode effect using a one-dimensional nonlinear spring-mass lattice with a light mass as a localised defect. A single light mass defect is embedded in an otherwise periodic chain of identical masses and springs. The position of the defect is varied to simulate both forward and reverse configurations. The system is excited harmonically at one boundary, and the resulting wave propagation is analyzed.

Eigenvalue analysis of the linearized system reveals a defect mode with a frequency above the acoustic band cutoff. When the system is driven at this defect frequency, amplitude-dependent behavior emerges. For low amplitudes, the defect mode remains localized and does not transmit energy. However, for high amplitudes, the system crosses a threshold and begins transmitting energy along the chain. In the forward configuration, this leads to significant wave transmission, whereas in the reverse configuration, transmission remains suppressed. This results in a diode-like transmission.

2 System description

The mechanical system analyzed in this study is a one-dimensional spring-mass chain. Nonlinearity of Hertzian contact forces¹ is replicated using nonlinear springs, which follow a cubic force-displacement relationship. The chain is excited at the left boundary, modeled as fixed, with a harmonic forcing. Two configurations are studied:

- *forward configuration* - the light mass defect at 2nd position from left boundary, i.e., driving near the defect.
- *reverse configuration* - the light mass defect at 2nd position from right boundary, i.e., driving far from the defect.

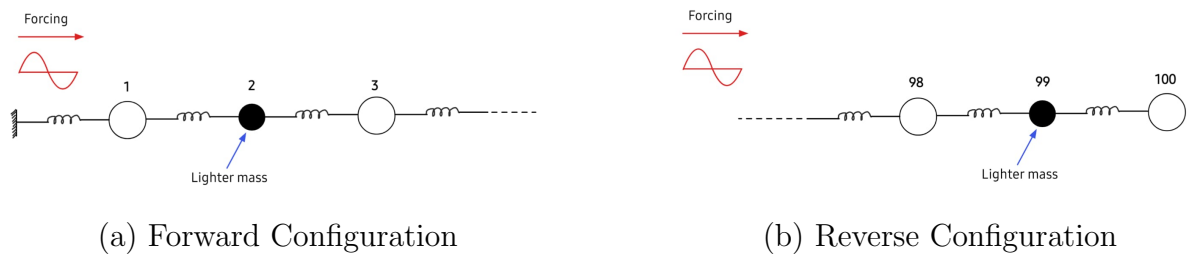


Figure 1: Comparison of two configurations

The numerical values used for analysis are:

$$\begin{aligned} N &= 100 \\ m &= 20 \text{ gm} \\ M &= 100 \text{ gm} \\ K_l &= 1500 \text{ N/m} \\ K_{nl} &= 1.5 \times 10^7 \text{ N/m} \end{aligned}$$

where N is the number of masses in the chain, m is the mass of the light mass defect, M is the mass of uniform masses, K_l and K_{nl} are stiffness of linear and nonlinear spring respectively.

3 Methodology

Equations of Motion

The equation for the n^{th} mass is written as:

$$M_n \ddot{u}_n = [K_l(u_{n-1} - u_n) + K_{nl}(u_{n-1} - u_n)^3] - [K_l(u_n - u_{n+1}) + K_{nl}(u_n - u_{n+1})^3]$$

where n varies from 2 to 99.

The equation of motion for the mass near the left (fixed) boundary is:

$$M \ddot{u}_1 = -[K_l u_1 + K_{nl} u_1^3] - [K_l(u_1 - u_2) + K_{nl}(u_1 - u_2)^3]$$

The equation of motion for the mass near the right (free) boundary is:

$$M \ddot{u}_{100} = [K_l(u_{99} - u_{100}) + K_{nl}(u_{99} - u_{100})^3]$$

where:

- u_n is the displacement of the n_{th} mass
- m_n is the mass at n_{th} index
- K_l and K_{nl} is the spring constants

Here, $M_2 = m$ in forward configuration and $M_{99} = m$ in reverse configuraton. The system of nonlinear differential equations is solved using *ode45*.

Energy transmitted and Transmission Ratio

A parameter *Transmission Ratio (T.R.)* is defined as the ratio of energy transmitted (to the last mass in the chain) in forward and reverse configuration.

$$T.R = \frac{\text{Energy Forward}}{\text{Energy Reverse}}$$

The energy transmitted to the last mass is estimated as the area under the PSD curve of its velocity.

4 Results and Discussion

Defect Mode Characterization

Eigenvalue analysis of the linearized system ($K_{nl} = 0$) is carried out. The single light mass defect generates a localized mode referred to as the defect mode. Its frequency ω_d is ω_c , where ω_c is the upper cutoff frequency of the acoustic band of the uniform spring mass system given by:

$$\omega_c = 2\pi\sqrt{\frac{K}{M}}$$

The values of these frequencies are:

$$\omega_c = 244.9176 \text{ rad/s} \quad \omega_d = 408.1374 \text{ rad/s}$$

Note that the defect mode frequency is not given by the expression used above, i.e.

$$\omega_d \neq 2\pi\sqrt{\frac{K}{m}}$$

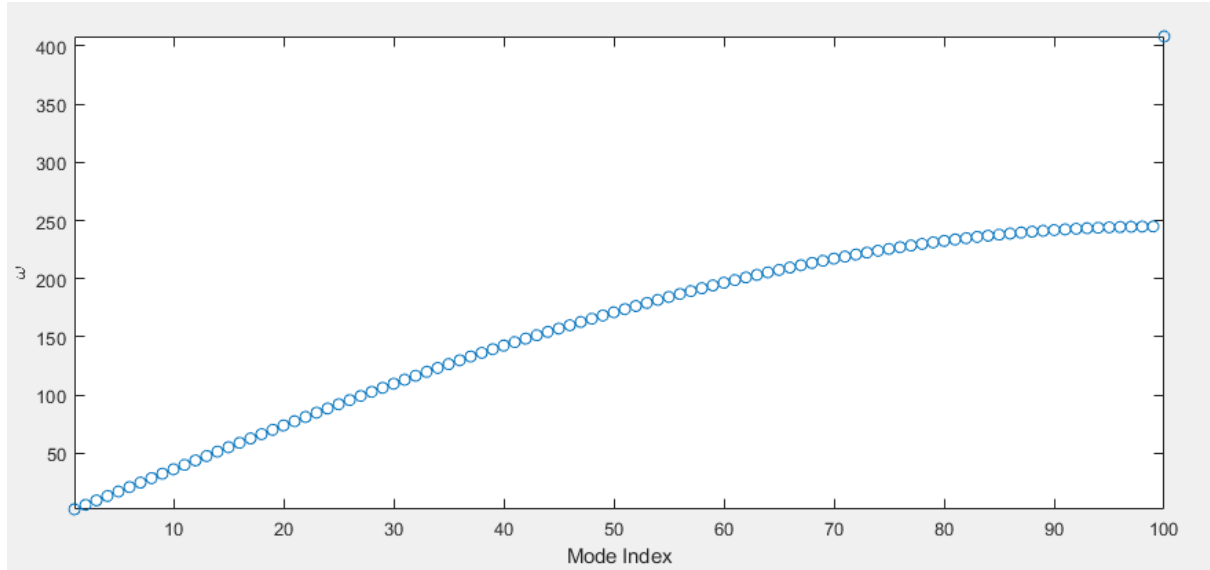


Figure 2: Dispersion Curve

The spatial profile of the defect mode shows adjacent particles oscillating out of phase.

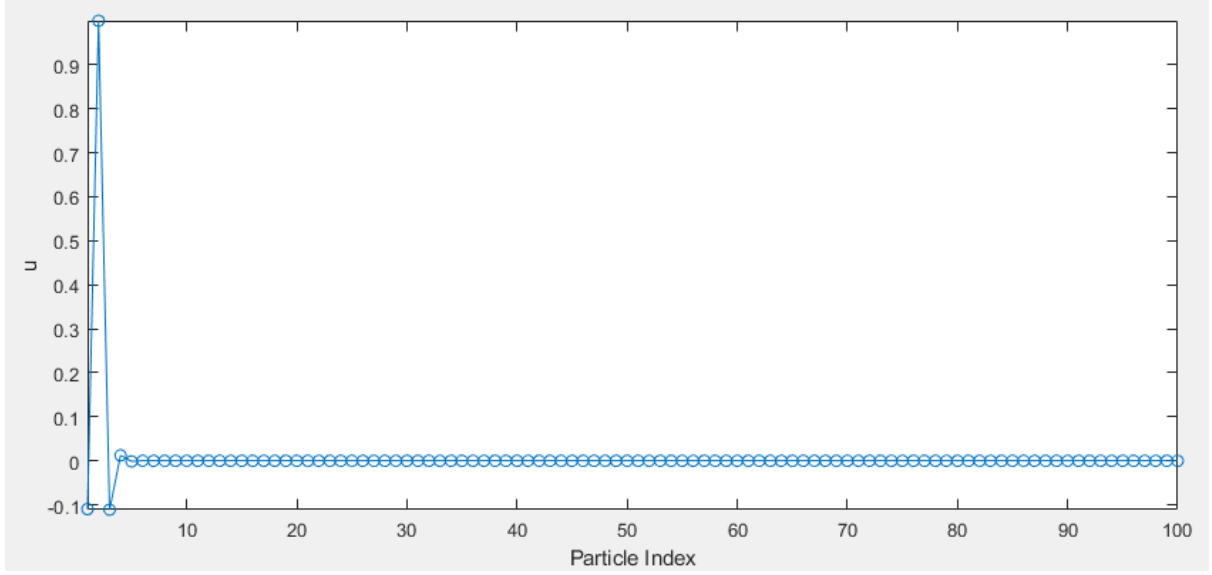


Figure 3: Spatial profile of the defect mode

Nonlinear Wave Propagation

The nonlinear chain in forward configuration is excited at ω_d with varying input amplitudes. From the spatiotemporal plot, threshold behavior is observed for wave propagation. For low amplitudes (≤ 10), the forcing excites a periodic vibrational mode localized around the defect. In this case, the energy does not propagate through the crystal. As the amplitude increases (≥ 100), wave propagation in the chain is observed, i.e., energy transmission starts. At higher amplitudes in the forward configuration, energy gets distributed to frequencies within the acoustic band.

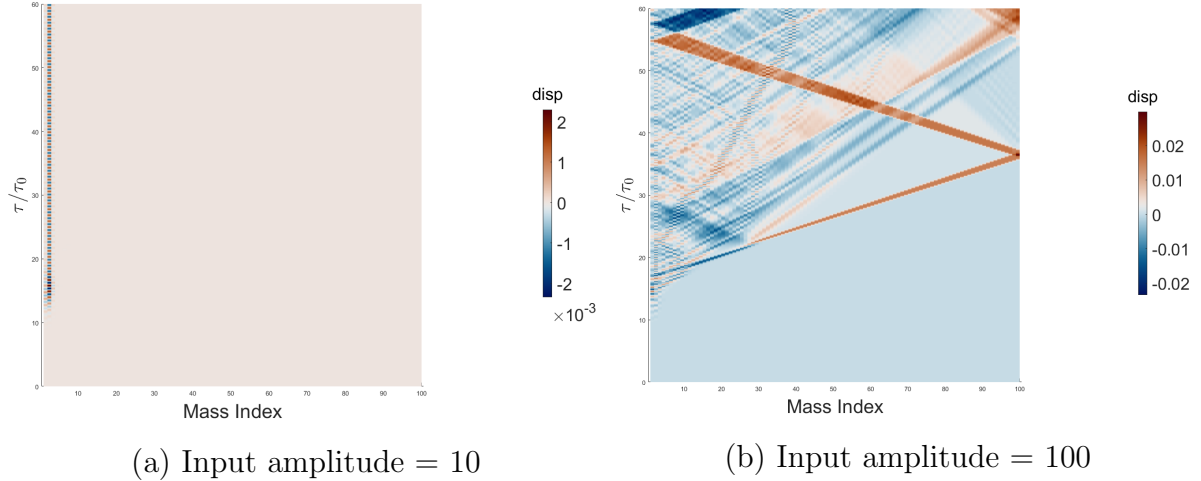


Figure 4: Threshold behavior due to nonlinearity (forward)

At the same amplitude, no wave propagation is seen in the reverse configuration. However, at high enough amplitudes, wave propagation happens here as well.

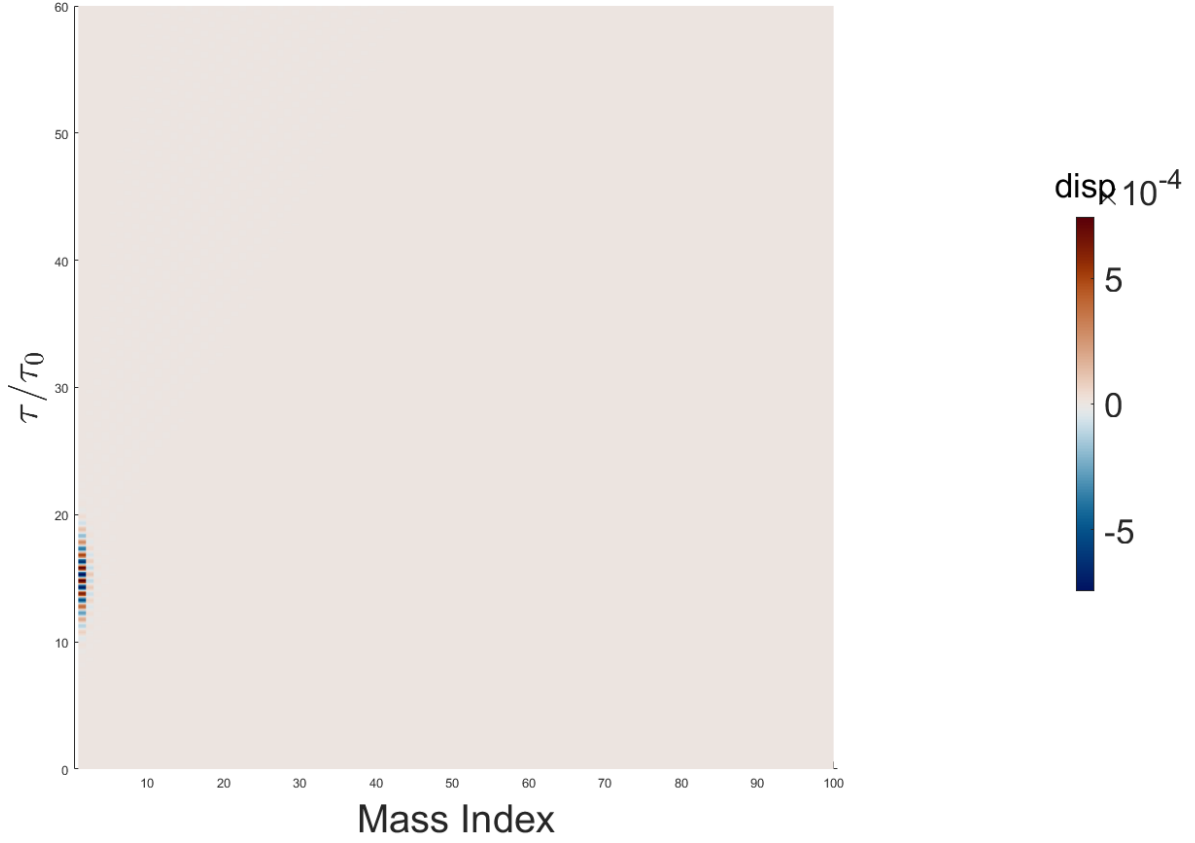


Figure 5: Input amplitude = 100 (reverse)

Defect Mode Characterization

The energy transmitted to the last mass is plotted against various input amplitudes. A sharp jump in energy transmission is seen in the forward configuration at the amplitude exhibiting threshold behavior. This is not true in the reverse case.

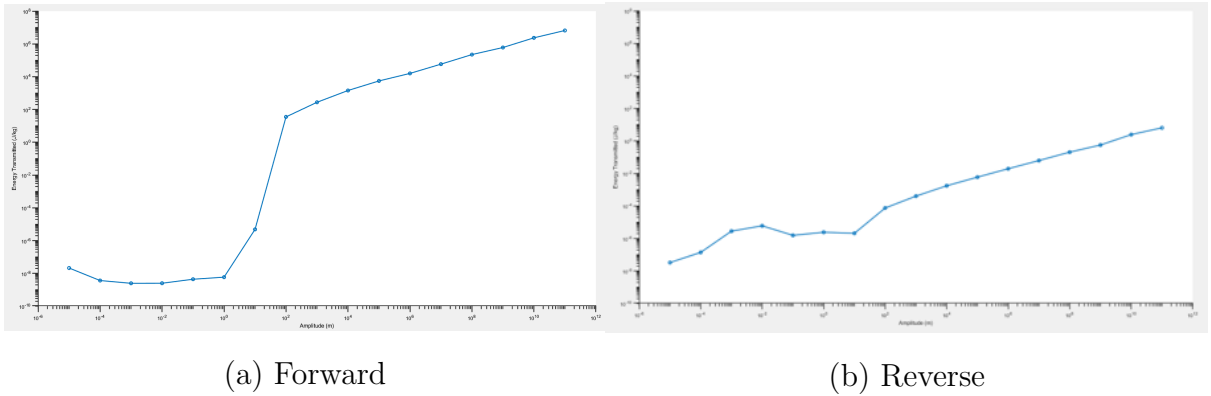


Figure 6: Energy vs input amplitude (at frequency $5Hz$ less than f_d)

The study is carried out at different frequencies near f_d . Within frequencies $10 Hz$ below f_d , $T.R$ is in the order of 10^5 .

At frequencies closer to f_c , energy transmission in reverse configuration increases significantly, thus reducing T.R..

Heavy Mass Defect

For $m = 1000$ kg, the defect node frequency falls in the acoustic band, i.e., the chain behaves like a monoatomic lattice.

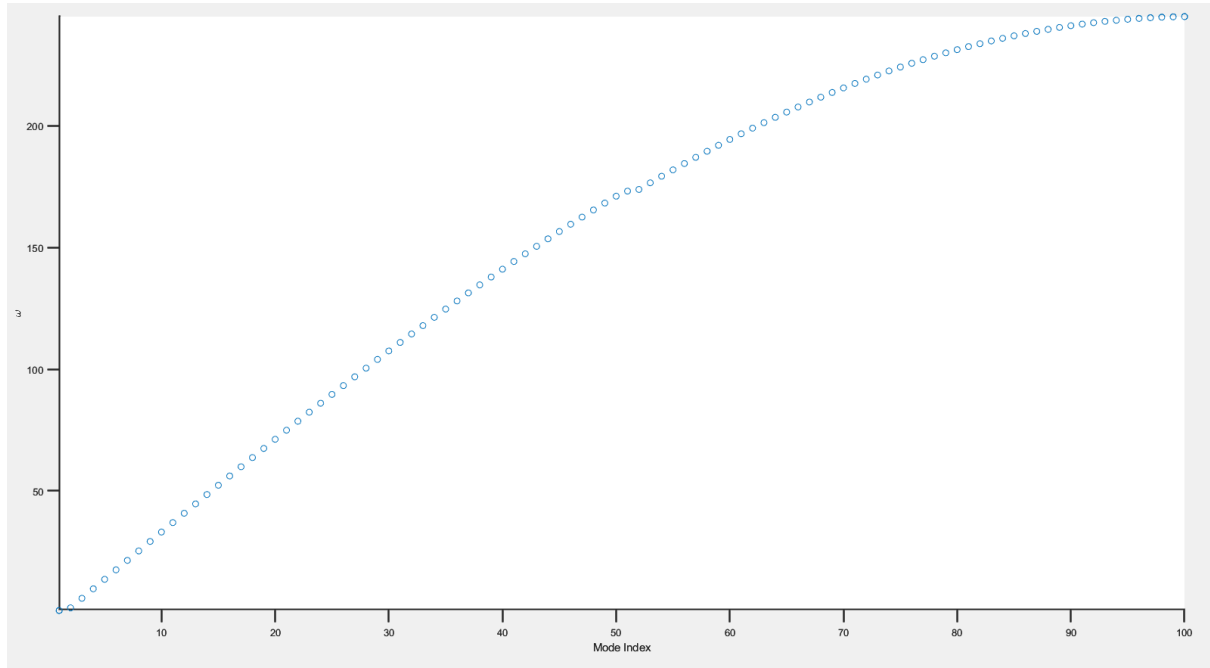


Figure 7: Dispersion Curve

5 Conclusion and Future Work

Conclusion

- The study modeled a nonlinear spring mass lattice with a light mass defect and demonstrated non-reciprocal wave propagation, mimicking the behavior of an acoustic diode.
- The light mass defect introduced a defect mode above the acoustic band.
- Nonlinear wave propagation showed amplitude-dependent threshold behavior.
- Diode effect observed with a transmission ratio ($T.R.$) $\simeq 10^5$ and an operational bandwidth of $10Hz$ below defect mode frequency.

Future Work

- Multiple or distributed defects at different positions could be used to engineer multiple localized modes. Interaction of these modes under nonlinearity may lead to richer diode functionalities.
- Designing experiments to validate the numerical results obtained in the study..
- Inclusion of damping to simulate realistic materials.

- Exploration of application-driven extensions of this study. Improved control over the transmission threshold, directionality, and operational bandwidth can support the development of wave-based logic gates, mechanical signal filters for noise control and vibration isolation, sensors, and switches, etc.

References

- Boechler N., Theocharis G., Daraio C., Bifurcation-based acoustic switching and rectification, Nature Materials 10, (2019) 665-668.
- Boechler N., Theocharis G., Man Y., Kevrekidis P. G. & Daraio C, Defect modes in one-dimensional granular crystals (2012). <http://arxiv.org/abs/1103.22483>.

Appendix: Matlab Code

```

1 N = 100;
2 m1 = 100e-3; m2 = 20e-3;
3 k = 1500;
4
5 mass = m1*ones(N,1);
6 mass(1+1, 1) = m2;           % defect at position 2
7
8 %% Linearized System Eigenvalue Analysis
9
10 M = diag(mass);
11 K = zeros(N, N);
12
13
14 for i = 1:N-1
15     K(i,i) = K(i,i) + k;
16     K(i+1,i+1) = K(i+1,i+1) + k;
17     K(i,i+1) = K(i,i+1) - k;
18     K(i+1,i) = K(i+1,i) - k;
19 end
20
21 K(1, 1) = K(1, 1) + k;
22
23 H = M\K;
24 [Phi, Lambda] = eig(H);
25 w = sqrt(diag(Lambda));
26 freq = w / (2*pi);
27
28
29 [sorted_freq, idx] = sort(freq);
30 Phi_sorted = Phi(:, idx);
31
32
33 mshapes = zeros(N, N);

```

```

34 for l = 1:N
35     mshapes(:, l) = Phi(:, idx(l));
36 end
37
38 figure(1);
39 plot(sorted_freq*(2*pi), 'o');
40 grid off;
41 xlabel('Mode_Index');
42 ylabel('\omega');
43 axis tight;
44 grid off;
45 ax = gca;
46 ax.Box = 'off';
47 ax.TickDir = 'out';
48 ax.LineWidth = 1.5;
49 %%
50 figure(2);
51 plot(-mshapes(:, end)/max(abs(mshapes(:, end))), 'o-');
52 grid off;
53 xlabel('Particle_Index');
54 ylabel('u');
55 grid off;
56 ax = gca;
57 ax.Box = 'off';
58 ax.TickDir = 'out';
59 ax.LineWidth = 1.5;

```

Listing 1: Matlab Code for Eigenvalue Analysis

```

1 clear; clc; close all;
2 tic;
3
4 load("vik.mat");
5 load("davos.mat");
6
7
8 N = 100;
9 m1 = 100e-3; m2 = 20e-3;
10 k = 1500; % linear spring stiffness
11 alpha = k*1e5; % cubic nonlinearity coefficient
12
13 mass = m1*ones(N,1);
14 mass(2, 1) = m2; % defect at position 2
15
16 % mass = flip(mass) % uncomment for reverse configuration
17
18
19 omega_defect = 408.1374;
20 omega_cutoff = 244.9176;
21 omega = omega_defect - 5*pi*2;
22 disp('omega_ = '); disp(omega);
23 amp_values = [1e-5, 1e-4, 1e-3, 1e-2, 1e-1, 1, 1e1, 1e2, 1e3, 1e4, 1e5, 1e6, 1

```

```

    e7, 1e8, 1e9, 1e10, 1e11];
24 t0 = 2*pi/omega;
25 dt = t0/10;
26 T = 15*t0;
27 tspan = 0:dt:T;
28
29 u0 = zeros(N,1);
30 v0 = zeros(N,1);
31 initial_conditions = [u0; v0];
32
33 t_c = 5 * t0;
34 sigma = 2 * t0;
35
36 for j = 1:length(amp_values)
37     drive = @(t) amp_values(j) * exp(-(t - t_c).^2 / (2 * sigma^2)) .* sin(
        omega * t);
38     ode_fun = @(t, y) odefunc(t, y, k, alpha, mass, N, drive);
39     [t, Y] = ode45(ode_fun, tspan, initial_conditions);
40     u_sol = Y(:, 1:N);
41     v_sol_N = Y(:, 2*N);
42     [ParticleIdxGrid, TimeGrid] = meshgrid(1:size(u_sol,2), t);
43     figure;
44     surf(ParticleIdxGrid', TimeGrid'/t0, u_sol', 'EdgeColor', 'none');
45     grid off;
46     xlabel('Mass_Index','FontName', 'Helvetica', 'FontSize', 30);
47     ylabel('$\tau_0$', 'FontName', 'Helvetica', 'FontSize', 30,'
        Interpreter','latex');
48     zlabel('Displacement','FontName', 'Helvetica', 'FontSize', 30);
49     title(sprintf('Forward(A=%f)', amp_values(j)));
50     view(-45,30);
51
52     a=daspect;
53     daspect([a(1) a(2) 4*a(3)]);
54     pbaspect([1 1 1]);
55
56     colormap(vik)
57     c = colorbar;
58     c.Location = 'east';
59     c.AxisLocation = 'out';
60     c.Position = [0.85 0.3 0.01 0.4];
61     set(c, 'FontName', 'Helvetica', 'FontSize', 26)
62     c.Title.String = 'disp';
63     v_lim=max(max(u_sol));
64     set(gcf,'color','w');
65
66     [F, XF] = myfft(t, v_sol_N);
67     PSD = XF;
68     dF = F(2) - F(1);
69     Area_PSD(j) = sum(PSD) * dF;
70 end
71

```

```

72 %%
73 figure;
74 loglog(amp_values, Area_PSD, 'o-', 'LineWidth', 2);
75 xlabel('Amplitude (m)');
76 ylabel('Energy Transmitted (J/kg)');
77 % title(sprintf('Energy vs Input Amplitude (Forward, alpha = %d)', alpha));
78 grid off;
79 ax = gca;
80 ax.Box = 'off';
81 ax.TickDir = 'out';
82 ax.LineWidth = 1.5;
83 toc;
84 %%
85
86 function dydt = odefunc(t, y, k, alpha, mass, N, drive)
87     % y = [u1, u2, ..., uN, v1, v2, ..., vN]
88     u = y(1:N);
89     v = y(N+1:end);
90     F = zeros(N,1);
91
92     % Internal masses (2 to N-1): force from both left and right neighbors
93     for i = 2:N-1
94         dxL = u(i) - u(i - 1);
95         dxR = u(i + 1) - u(i);
96
97         FL = k * dxL + alpha * dxL^3;
98         FR = k * dxR + alpha * dxR^3;
99
100         F(i, 1) = -FL + FR;
101     end
102
103     % Mass 1 (leftmost): connected to the wall through a spring
104     F(1, 1) = - (k * u(1) + alpha * u(1)^3) - (k * (u(1) - u(2)) + alpha * (u(1) - u(2))^3);
105
106     % Mass N (rightmost): free boundary
107     F(N, 1) = k * (u(N - 1) - u(N)) + alpha * (u(N - 1) - u(N))^3; % No spring at free end
108
109     % External drive:
110     F(1, 1) = F(1, 1) + drive(t);
111
112     % System of ODEs
113     du_dt = v;
114     dv_dt = F ./ mass;
115
116     dydt = [du_dt; dv_dt];
117 end
118
119 function [F, Sxx] = myfft(T, X)
120

```

```

121 Fs = 1 / (T(2) - T(1));
122 dt = T(2) - T(1);
123 nfft = 2^nextpow2(length(T));
124
125 % Compute FFT
126 Xfft = fft(X, nfft, 1);
127
128 % Two-sided PSD
129 Sxx2 = (abs(Xfft).^2) / (Fs * nfft);
130
131 % One-sided PSD
132 Sxx = Sxx2(1:nfft/2+1, :);
133 Sxx(2:end-1,:) = 2 * Sxx(2:end-1,:);
134
135 % Frequency vector (in Hz)
136 F = (0:(Fs/nfft):(Fs/2));
137
138 end

```

Listing 2: MATLAB Code for Forward and Reverse Configuration